SDXL LoRA + DreamBooth: Fine-Tuning Text-to-Image Diffusion Models for Subject-Driven Generation

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Declaration

We do hereby declare that the research works presented in this thesis entitled, " SDXL LoRA + DreamBooth: Fine-Tuning Text-to-Image Diffusion Models for Subject-Driven Generation" are the results of our own works. We further declare that the thesis has been compiled and written by us and no part of this thesis has been submitted elsewhere for the requirements of any degree, award or diploma, or any other purposes except for publications. The materials that are obtained from other sources are duly acknowledged in this thesis.

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Approval

We hereby declare that the research works presented in this thesis entitled, " SDXL LoRA + DreamBooth: Fine-Tuning Text-to-Image Diffusion Models for Subject-Driven Generation" is the result of the original work carried out by Syed Rubad, Md Rayhan, A. K. M. Jahidul Islam Jewel, Miah Mohaiminur Rahman, and Md Shahriar Alam under my supervision. I further declare that no part of this thesis has been submitted elsewhere for the requirements of any degree, award diploma, or any other purposes except for publications. I further certify that the dissertation meets the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

”Dedicated to the pursuit of knowledge and to the countless individuals who have illuminated our path with their guidance, encouragement, and unwavering belief in the power of research.”

Acknowledgement

This research journey has been a profound learning experience, and we owe our gratitude to many individuals and our institution who made it possible. We extend our deepest appreciation to our dedicated supervisor, T. M. Amir - Ul - Haque Bhuiyan, Assistant Professor, Department of CSE, BUBT, whose insightful guidance, unwavering support, and mentorship were pivotal in shaping the course of this research. Your expertise and belief in my work were instrumental in achieving the goals of this thesis. We would like to acknowledge the faculty and staff at Bangladesh University of Business and Technology (BUBT), who provided an environment of academic excellence, resources, and opportunities for intellectual growth. Our heartfelt gratitude goes to our family for their boundless encouragement and love. Your enduring support and belief in our abilities were the driving forces that kept us motivated. To our friends and teammates, we are grateful for the stimulating discussions, exchange of ideas, and camaraderie that made this academic journey enjoyable. This work is grounded in the foundational research of the generative artificial intelligence community, particularly in the areas of text-to-image diffusion models, Ensemble in text generation, LoRA-based fine-tuning, and DreamBooth-driven personalization. While individual contributions cannot all be explicitly acknowledged, their collective efforts have significantly influenced this study. We sincerely appreciate these contributions and dedicate this work to advancing subject-driven image generation through rigorous scientific research.

Abstract

Recent advancements in diffusion-based text-to-image generation have enabled models to synthesize visually rich and detailed images directly from natural language descriptions. Although these models achieve impressive image quality, consistently preserving the visual identity of a specific subject across varying styles, scenes, and viewpoints remains a challenging problem. To address this limitation, this paper introduces a subject-centric text-to-image generation framework that integrates ensemble text prompts for generating specific subjects with the DreamBooth technique combined with Low-Rank Adaptation (LoRA) on top of the Stable Diffusion XL (SDXL) architecture. The proposed framework personalizes a pretrained SDXL model using a limited number of subject images by binding the subject's identity to a dedicated textual token. In contrast to traditional DreamBooth training, which involves full model fine-tuning and explicit class-based prior preservation, our approach leverages LoRA-based fine-tuning while keeping the underlying diffusion model parameters fixed. This design significantly reduces computational requirements, enhances training stability, and implicitly maintains class-level semantic knowledge. Furthermore, the framework provides increased flexibility during inference, enabling the personalized subject to be naturally embedded into unseen scenes, artistic styles, and compositions through diverse text prompts. The adoption of LoRA adapters also facilitates efficient storage and reuse of subject-specific information without the need to replicate the entire diffusion model. Experimental evaluations demonstrate that the proposed approach successfully generates identity-consistent images of the target subject under a wide range of prompts, visual styles, and environmental conditions, underscoring the effectiveness and practical utility of the SDXL LoRA–DreamBooth framework for efficient subject-driven image generation.

Keywords: Text-to-Image Generation, Stable Diffusion XL (SDXL), Ensemble, DreamBooth, Low-Rank Adaptation (LoRA), Diffusion Models, Subject-Driven Image Generation, Personalized Image Synthesis

List of Abbreviations and Acronyms

AI	Artificial Intelligence
CNN	Convolutional Neural Network
CV	Computer Vision
DDPM	Denoising Diffusion Probabilistic Model
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
ICLR	International Conference on Learning Representations
ICCV	International Conference on Computer Vision
LoRA	Low-Rank Adaptation
NeurIPS	Conference on Neural Information Processing Systems
SD	Stable Diffusion
SDXL	Stable Diffusion XL
UNet	U-shaped Convolutional Neural Network
VAE	Variational Autoencoder

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Chapter 1

1 Introduction

1.1 Introduction

Artificial intelligence (AI) has become a transformative force in a variety of fields, fundamentally changing how digital systems perceive, create, and interact with visual information. One of the fastest-growing areas within AI is generative modeling, particularly text-to-image generation. Text-to-image generation refers to the task of synthesizing images directly from natural language descriptions, enabling machines to translate abstract textual concepts into visually meaningful representations. This capability has gained widespread popularity due to its applications in creative design, advertising, entertainment, education, virtual reality, and human-computer interaction. In the early stages of computer vision research, image generation relied on deterministic and rule based approaches that required extensive manual design and domain-specific knowledge. These approaches were limited in scalability and struggled to generate complex or realistic images. The introduction of deep learning has marked a significant paradigm shift, enabling models to automatically learn visual representations from large datasets. Among these methods, generative adversarial networks (GANs) emerged as a breakthrough, training two competing networks – a generator and a discriminator – to produce visually realistic images. Despite their success, GAN-based models exhibited several limitations, including training instability, mode collapse, and difficulty in producing consistent results [1]. These limitations motivated the exploration of alternative generative frameworks, leading to the emergence of diffusion-based models. Recent studies have shown that diffusion-based generative models achieve higher image quality and training stability than GAN-based methods by iteratively transforming random noise into meaningful images [1, 2]. This method has demonstrated higher stability, diversity, and image quality than GAN-based methods [1] furthermore, this study explores the use of

ensemble techniques.

Stable Diffusion and its improved variant, Stable Diffusion XL (SDXL), represent state-of-the-art latent diffusion models capable of generating high-resolution and, therefore, accurate images [3]. Latent diffusion models such as Stable Diffusion significantly reduce computational costs by performing the diffusion process in a compressed latent space, while Stable Diffusion XL (SDXL) further improves the immediate intelligibility and visual fidelity through architectural enhancements [3, 4]. However, these models are trained on large-scale datasets and primarily produce generic representations of concepts rather than preserving the identity of a specific subject. This limitation forms the main motivation for this research, which aims to enable personalized text-to-image generation using SDXL in combination with DreamBooth and LoRA.

1.2 Problem Statement

Although modern diffusion-based text-to-image models demonstrate remarkable capabilities, they face a fundamental limitation when applied to personalized image generation. In particular, these models are unable to consistently generate images of a specific subject across different prompts and contexts. When given a text description of a subject, such as a specific car or object, the model generates visually plausible images; however, each generated image often represents a different instance rather than the same subject. This inconsistency poses a major challenge in applications where subject identity is important. For example, in product visualization, marketing campaigns, character design, and personalized creative content, it is essential to maintain the same visual identity across multiple images.

This limitation arises because pre-trained diffusion models are optimized to learn a distribution-level representation of a specific subject across different prompts rather than preserving the instance-level identity of a specific subject [5]. Pre-trained diffusion models lack a mechanism to associate a unique subject identity with a text token unless explicitly trained to do so. Traditional approaches to solving this problem include retraining or fine tuning the entire model on subject-specific data.

However, such methods are computationally expensive, require large memory resources, and are impractical for most academic and real-world users. Moreover, complete fine-tuning increases the risk of overfitting and catastrophic forgetting, where the model loses its generalization ability. Therefore, to address these challenges, ensemble-based strategies—such as the fusion of multiple Low-Rank Adaptation (LoRA) weights or the concurrent use of specialized model checkpoints—can be employed to stabilize identity preservation while maintaining the model's original creative flexibility.

1.3 Problem Background

The background to this problem lies in the evolution of generative image synthesis techniques. Early generative models relied on probabilistic frameworks such as variational autoencoders (VAEs), which introduced latent representations but often produced fuzzy outputs. GANs significantly improved realism but suffered from training difficulties and instability, especially for high-resolution images. Diffusion models emerged as a powerful alternative by modeling the image generation process as a sequence of noise removal steps. Latent diffusion models further increased efficiency by operating in a compressed latent space rather than pixel space. Stable Diffusion XL extends this framework by incorporating a larger UNET architecture and dual text encoders, enabling better understanding of complex prompts and improved visual fidelity. Despite these advances, diffusion models are fundamentally trained to capture distribution-level patterns rather than instance-level identities. This means that they are good at producing images that match a category or concept but fail to consistently reproduce a specific instance. Dreambooth was introduced to address this limitation by fine-tuning a unique identifier token to a subject. However, the original Dreambooth approach requires fine-tuning of the full model and explicit prior preservation loss, which increases the complexity of training. This research builds on the Dreambooth concept while addressing its limitations through the integration of LoRA and SDXL.

1.4 Research Objectives

The primary objective of this research is to develop a subject-driven text-to-image generation framework that enables identity-preserving image synthesis using a pre-trained diffusion model. This study focuses on personalizing a powerful generative model while maintaining visual consistency of the target subject across diverse prompts, with an emphasis on efficiency and qualitative performance.

The specific objectives of this research are as follows:

- Investigate the effectiveness of Stable Diffusion XL (SDXL) as a base model for subject personalization tasks, and examine the usefulness of an ensemble approach in generating accurate and subject-specific image prompts.
- Apply the DreamBooth method to associate a unique identifier token with a target subject using a limited number of reference images.
- Explore the use of Low-Rank Adaptation (LoRA) as a parameter-efficient fine-tuning technique to reduce computational cost and memory requirements.
- Evaluate the qualitative performance of the personalized model in generating consistent and visually diverse images under different text prompts.
- Analyze the similarities and differences between the proposed framework and the original DreamBooth approach in terms of performance, efficiency, and flexibility.

1.5 Motivation

The motivation behind this research stems from the growing demand for personalized artificial intelligence (AI)-generated content in both academia and industry. While large scale distributed models are powerful, their deployment is often constrained by computational constraints, making personalization inaccessible to many users. Using LoRA, this research attempts to democratize personalized text-to-image generation by enabling efficient fine-tuning on limited hardware. The motivation is driven by the desire to reduce training complexity while maintaining higher-quality and identity-consistent output.

1.6 Flow of Research

The research follows a systematic workflow that starts with an extensive literature review on generative models and personalization techniques. This is followed by the selection of SDXL as the base model and the integration of DreamBooth principles using LoRA. Subsequently, subject images are collected and used to fine-tune the model. The trained model is then evaluated using various text prompts to assess identity preservation and visual quality. Finally, the results are analyzed and documented.

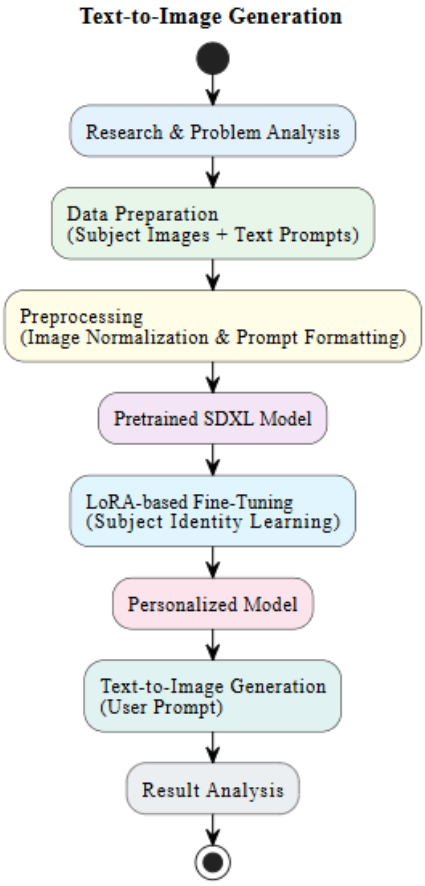

Figure 1.1: Flow Diagram of the Thesis Work

1.7 Significance of the Research

The significance of this research is its contribution to the practical advancement of personalized text-to-image generation using diffusion-based models. While state-of-

the art models such as Stable Diffusion XL demonstrate exceptional performance in generating high-quality images from text prompts, their inability to preserve the identity of a specific subject significantly limits their applicability in real-world situations. This research addresses that limitation by introducing an efficient and scalable subject personalization framework that enables the generation of identity consistent images without extensive computational resources as well as unified best prompt through ensemble. One of the most important aspects of this research is its focus on accessibility. Traditional personalization techniques with full model fine-tuning demand high-end GPUs, long-training times, and large memory capacity, which are often unavailable in academic environments or small industries. By integrating low-rank adaptation (LoRA) into the DreamBooth framework, this research significantly reduces the training-complexity while maintaining high-quality output. This makes personalized diffusion models possible for a wide range of users, including students, researchers, and independent developers. Furthermore, the study contributes to the growing body of work on efficient adaptation of large pre-trained models. Instead of changing millions of parameters, the proposed method only updates a small set of trainable parameters, thereby preserving the generalization ability of the base SDXL model. This implicit preservation of class-level knowledge ensures that the personalized model remains versatile and is able to generate diverse images beyond the training examples. In addition, this study is significant for various application domains, including digital marketing, product visualization, creative arts, and virtual environment design. The ability to generate consistent images of a specific subject under different prompts increases the reliability and user control of the content. From an academic perspective, the study provides a concrete implementation and analysis of Dreambooth-style personalization using SDXL, providing valuable insights for future studies in generative modeling and model adaptation.

1.8 Research Contribution

This study advances personalized text-to-image generation by combining subject-driven learning, efficient fine-tuning, and smarter prompt design. The result is

higher-quality, more controllable personalized images — with much lower computational cost.

Our main contributions are:

- **DreamBooth-style personalization in SDXL**

  We successfully adapt DreamBooth-style subject-driven generation to Stable Diffusion XL (SDXL), taking advantage of its stronger architecture for better prompt understanding and sharper, more faithful images.

- **Ensemble-based prompt generation**

  Instead of a single prompt, we generate multiple diverse prompts and combine them to guide the diffusion process. This improves text-image alignment, reduces sensitivity to wording, avoids mode collapse, and gives more consistent and controllable results.

- **Parameter-efficient fine-tuning with LoRA**

  We use Low-Rank Adaptation (LoRA) to personalize the model efficiently: only small low-rank updates are trained and added to attention layers. This cuts memory use and training time dramatically while keeping the process stable.

- **Simple prior preservation via parameter freezing**

  Rather than using an extra class-specific loss (like original DreamBooth), we freeze most of the base SDXL weights. This naturally preserves general knowledge, avoids overfitting, and keeps the model flexible for varied realistic outputs.

- **Qualitative study of identity preservation**

  Through detailed visual analysis across many prompts, scenes, styles, and angles, we show that our LoRA-personalized model reliably keeps the subject's identity , providing strong evidence of its effectiveness in large diffusion models.

1.9 Thesis Organization

This thesis is divided into six chapters, each focusing on a specific aspect of the research to ensure clarity and logical progression. Chapter 1 introduces the re-

search topic, outlines the problem statement, presents the objectives and motivation, and discusses the significance and contributions of the research. Chapter 2 provides a comprehensive review of the existing literature on image generation, generative adversarial networks, diffusion models, ensemble models, stable diffusion, SDXL, Dreambooth, and low-rank adaptation. This chapter critically analyzes previous work to identify research gaps and validate the proposed approach. Chapter 3 presents the proposed model and research methodology in detail. It describes the system architecture, feasibility analysis, dataset preparation, and integration of Dreambooth and LoRA within the SDXL framework. The chapter also explains the design and implementation strategy adopted in this study. Chapter 4 focuses on the implementation and testing stages. It discusses the dataset usage, training methodology, evaluation strategy, and results analysis. Qualitative evaluation is presented to demonstrate identity preservation and image quality. Chapter 5 highlights the standards, limitations, policy considerations, and project management aspects, including challenges, constraints, and timeline analysis. This chapter ensures that the research is consistent with academic, technical, and policy standards. Finally, Chapter 6 concludes the thesis by summarizing the main findings and contributions. It also discusses possible future research directions and improvements of the proposed system.

1.10 Summary

This chapter presents a comprehensive introduction to research on personalized text-to image generation using Stable Diffusion XL (SDXL), which is integrated with Dreambooth and Low-Rank Adaptation (LORA). It reviews the background and evolution of generative image synthesis, with particular attention to recent advances in diffusion-based models and subject-driven personalization techniques. The chapter identifies the fundamental challenge of preserving subject identity across different visual contexts, styles, and perspectives, which remains a key limitation in existing text-to-image generation frameworks. Furthermore, the motivation and objectives of this research are clearly articulated, highlighting the need for an efficient

and scalable personalization approach that balances image quality, identity consistency, and computational cost. The significance and contributions of the proposed framework are discussed, emphasizing both its practical applicability to real-world creative systems and its academic relevance to ongoing research in generative artificial intelligence. Finally, the chapter concludes with a summary of the overall thesis structure, outlining the content and focus of the subsequent chapters, and laying a solid foundation for the subsequent detailed technical and experimental analysis.

Chapter 2

2 Literature Review

2.1 Introduction

The rapid evolution of deep learning has significantly impacted the field of image generation, leading to the development of increasingly sophisticated generative models. The purpose of this chapter is to review existing research on text-to-image generation, diffusion models, and subject personalization techniques. By examining previous work, this chapter establishes the theoretical foundation for the proposed research and identifies the limitations of existing approaches that motivated the development of the proposed model. Text-to-image generation lies at the intersection of natural language processing and computer vision, requiring models to understand textual semantics and translate them into coherent visual representations. Early research focused on simple image synthesis tasks, but recent advances have enabled the generation of high-resolution, photorealistic images from complex textual prompts. This progress has been driven largely by improvements in neural network architectures, training methods, and the availability of large-scale datasets. Despite these advances, subject-level personalization remains a challenging problem. Most pre-trained generative models are optimized to generalize to a wide data distribution rather than a continuum of identities. As a result, it becomes difficult to generate a picture of a specific subject without retraining or fine-tuning the model. This chapter reviews key developments in generative modeling, including generative adversarial networks, diffusion-based models, stable diffusion and stable diffusion XL, DreamBooth, and low-rank adaptation. The strengths and limitations of each approach are discussed to highlight the research gaps addressed in this thesis. The chapter concludes with an analysis of a problem that directly inspires the proposed SDXL-based DreamBooth with LoRA framework.

2.2 Literature Review

Early generative image synthesis methods were primarily based on Generative Adversarial Networks (GANs). The introduction of conditional GANs enabled image generation conditioned on auxiliary information such as class labels or textual descriptions [6]. Variational autoencoders (VAEs) introduced the concept of learning latent representations for image synthesis; however, they often produced blurry output due to their reliance on pixel-based reconstruction losses. The introduction of generative adversarial networks (GANs) has been a significant advance, enabling the generation of sharp and realistic images through adversarial training. GAN-based models such as DCGAN, StyleGAN, and BigGAN have demonstrated impressive visual quality but have suffered from training instability, mode collapse, and sensitivity to hyperparameters. Although GAN-based models such as StyleGAN produce visually sharp images, they often suffer from training instability and mode collapse, which has motivated the adoption of diffusion-based generative models [1]. Beyond GAN-based approaches, large-scale autoregressive models demonstrated strong zero-shot text-to-image generation capabilities by jointly modeling text and image representations [7]. Despite the dominance of diffusion-based models, recent studies such as GigaGAN have shown that GANs can still achieve competitive image quality with significantly faster inference when scaled appropriately [8].

To overcome these limitations, diffusion models were introduced as a powerful alternative. Diffusion models create images by gradually reversing the noise addition process, converting random noise into structured images through iterative denoising steps. These models exhibit stable training behavior and higher diversity than GANs.Although early GAN-based text-to-image models were effective, they were limited to low-resolution image synthesis. To address this limitation, StackGAN introduced a multi-stage architecture that progressively generated high-resolution and photo-realistic images [9]. Latent diffusion models operate in a low-dimensional latent space, which enables high-resolution image synthesis with reduced computational requirements [3]. Stable diffusion is a prominent example of this approach,

producing high quality images with reduced computational requirements. Stable diffusion XL further extends this architecture by incorporating a larger UNet backbone, dual text encoders, and improved attention mechanisms, thereby improving prompt comprehension and image fidelity. While these models excel at generating generic images, they struggle with subject personalization. Textual inversion learns new token embeddings to represent personalized concepts; however, it often struggles to capture complex subject identities using limited parameters [10]. DreamBooth enables diffusion models to associate a unique identifier token with a specific subject through topic-driven image generation, which achieves strong identity preservation [11]. However, full model fine-tuning is computationally expensive and requires explicit prior preservation losses. Low-rank adaptation (LoRA) introduces trainable low-rank matrices into trained models, enabling efficient fine-tuning with significantly fewer parameters [12]. Recent studies have shown that LoRA can effectively personalize diffusion models and preserve generalization. This thesis builds on these results by integrating DreamBooth and LoRA within the SDXL framework.

While diffusion-based models are highly effective at generating generic images, they often face limitations when it comes to topic-level personalization. Textual inversion addresses this issue by learning new token embeddings for personalized concepts; however, due to its limited parameter capacity, it frequently struggles to capture complex and fine-grained topic identities [10]. In addition, relying on a single textual prompt makes the personalization process highly sensitive to prompt formulation, which can lead to inconsistent generation results [13]. To overcome these challenges, ensemble-based prompting has been proposed, where multiple semantically related prompts are jointly used to guide the generation process. This strategy provides richer textual supervision and helps improve topic consistency and identity preservation across generated images [14].

2.3 Problem Analysis

Analysis of the existing literature reveals a significant gap between high-quality image generation and effective subject personalization. Although diffusion models such as SDXL demonstrate remarkable ability to generate realistic images, they have no mechanism to consistently preserve the identity of a specific subject across different prompts. This limitation arises because these models are trained to learn a broad data distribution rather than an instance-level representation. GAN-based personalization methods often require large amounts of subject-specific data and encounter instability during training. Textual inversion, although computationally efficient, struggles to represent complex visual identities using token embeddings alone. Then Dreambooth improves subject fidelity with the help of ensemble but introduces new challenges, including high computational cost, risk of overfitting, and the need for explicit prior preservation techniques. Furthermore, hardware limitations make full fine-tuning of large diffusion models impractical for many academic and industrial settings. This limits the accessibility of personalized image generation technologies and limits their adoption in real-world applications. The literature also highlights the importance of preserving class-level knowledge during personalization. Without proper constraints, finely-tuned models may lose their ability to generate diverse images, leading to reduced generalization. While the loss of prior preservation solves this problem, they add complexity to the training process. These challenges motivate the need for an efficient, stable, and scalable personalization framework. The integration of LoRA with SDXL provides a promising solution by reducing training costs, improving stability, and indirectly preserving class-level priors. This research aims to fill the identified gap by implementing a practical topic-driven text-to-image generation system.

2.4 Summary

This chapter reviews the existing research on text-to-image generation and subject personalization. It discusses the evolution of generative models from GANs to diffu-

sion based architectures and highlights the strengths and limitations of each method. Special attention was paid to Ensemble, Stable Diffusion XL, Dreambooth, and low-rank adaptation to form the basis of the proposed research. The problem analysis identified key limitations of existing methods, including high computational cost, training instability, and insufficient identity preservation. These revelations justify the need for an efficient personalization framework to balance subject fidelity and generalization. The insights gained from this literature review directly inform the design of the proposed model, which is presented in the next chapter. It addresses the identified research gaps. Recent surveys highlight that efficient personalization of diffusion models remains an open challenge, especially in terms of balancing identity preservation, generalization, and computational efficiency [15, 16].

Chapter 3

3 Proposed Model

3.1 Introduction

This chapter presents the subject-driven text-to-image generation model proposed in this study. The proposed system is designed to personalize a large-scale pre-trained diffusion model so that it can consistently generate images of a specific subject while preserving visual identity across different text prompts. The model is built on the Stable Diffusion XL (SDXL) architecture and incorporates Dreambooth-style subject binding combined with Low-Rank Adaptation (LoRA) for efficient fine-tuning as well as ensemble for batter subject recognition . The motivation behind the proposed model is to overcome the limitations identified in existing personalization methods. While the original Dreambooth achieves strong identity preservation, it requires full model fine tuning and explicit pre-preservation loss, resulting in high computational cost and increased training complexity. In contrast, the proposed model freezes the base SDXL parameters and introduces LoRA adapters to selectively update the attention layers, which significantly reduces memory usage and training time. This chapter discusses the feasibility, requirements analysis, and detailed research methodology of the proposed system. It further explains the architectural components of Ensemble, SDXL, and the role of DreamBooth in subject personalization, and the integration of LoRA for parameter-efficient training. Dataset preparation, system design, and implementation techniques are also presented to provide a complete understanding of the proposed framework.

3.2 Feasibility Analysis

Feasibility analysis is a crucial step that assesses whether a proposed system can be realistically designed, implemented, and deployed within specified constraints. This study conducts a comprehensive feasibility analysis to evaluate the technical,

economic, operational, and scheduling aspects of the proposed SDXL-based personalization framework. The feasibility analysis confirms that the proposed approach is not only theoretically sound but also practically implementable using available resources. Considering the scale of modern diffusion models, careful consideration of hardware requirements, training costs, and usability is essential. By adopting LoRA-based fine tuning, the proposed system aims to balance performance and efficiency, making subject driven image creation achievable in educational and resource-limited environments. The following subsections discuss in detail the different levels of feasibility.

3.2.1 Technical Feasibility

The technical feasibility tests whether the proposed system can be implemented using existing technologies and infrastructure. Stable Diffusion XL is an open-source latent diffusion model with a documented architecture and community support, making it technically viable for customization and experimentation. The availability of pre-trained SDXL checkpoints significantly reduces the need for extensive training from scratch. The integration of Dreambooth-style personalization is technically feasible due to SDXL's support for text conditioning via a cross-attention mechanism. The use of unique identifier tokens allows the model to bind a subject's identity to textual prompts. By combining Dreambooth-style subject binding with LoRA-based fine-tuning, the proposed framework enables identity-preserving personalization while keeping the base diffusion model frozen [11, 12]. Furthermore, Low-Rank Adaptation provides a proven mechanism for efficiently fine tuning large models by introducing a small number of trainable parameters at the attention level. From a hardware perspective, LoRA significantly reduces GPU memory requirements compared to full fine-tuning. As a result, the proposed system can be trained using consumer-grade GPUs or cloud-based platforms, making it technically feasible for academic research projects.

3.2.2 Economic feasibility

Economic feasibility assesses the cost-effectiveness of the proposed system. The complete fine-tuning of diffusion models such as SDXL requires considerable computational resources, resulting in high operational costs. These costs often make personalized model training inaccessible to students and small research teams. By employing LoRA-based fine-tuning, the proposed model reduces computational costs by training a fraction of the total parameters. This reduction in training costs makes the system economically viable for academic institutions with limited budgets. In addition, the use of open-source tools and publicly available pre-trained models further reduces development costs. The proposed approach eliminates the need for large-scale datasets and expensive data collection processes, as subject personalization can be achieved using a small number of images. This further increases the economic feasibility of the system.

3.2.3 Operational feasibility

The functional feasibility focuses on the usability and maintainability of the proposed system. The proposed framework is designed as modular, allowing different components such as dataset preparation, training, and inference to be handled independently. This modularity simplifies system management and debugging. The training process follows a straightforward workflow, requiring minimal manual intervention after defining the dataset and configuration parameters. During inference, users can create personalized images by including unique identifier tokens in text prompts. This user-friendly interaction makes the system functionally feasible for non-expert users. Furthermore, the frozen base model ensures stability during training, reducing the risk of unexpected behavior or performance degradation. This increases the reliability and maintainability of the system.

3.2.4 Scheduling feasibility

Feasibility is determined by determining whether the project can be completed within the specified timeframe. The proposed method significantly reduces training time compared to the full fine-tuning approach. LoRA-based personalization typically requires fewer training iterations, enabling faster experimentation and evaluation. The research workflow - literature review, model design, dataset preparation, training, evaluation, and documentation - was carefully planned to ensure timely completion. The reduced computational burden allows multiple experiments to be conducted within a limited timeframe, making the project schedule feasible for an academic defense.

3.3 Requirement Analysis

Requirements analysis has been performed to identify the hardware, software, and data requirements required to implement the proposed model. From a hardware perspective, the system requires a GPU capable of handling latent diffusion models, although memory requirements are significantly reduced due to LoRA-based fine-tuning. Software requirements include a deep learning framework such as PyTorch, diffusion model libraries, and tokenizer support for handling custom identifier tokens. Additional tools are required for image preprocessing and result visualization. Data requirements are minimal compared to traditional deep learning approaches. The system only requires a small set of subject images, typically ranging from 5 to 20 images, captured at different angles and lighting conditions. This minimal data requirement makes the system practical and scalable.

3.4 The Research Methodology

The research methodology follows a structured approach to ensure reproducibility and transparency. The process begins by selecting a pre-trained SDXL model as the base architecture. A unique identifier token is introduced to represent the target subject, and the identifier token is combined with the subject class name to create

training prompts. The LoRA adapter is then inserted into the selected attention level of the SDXL model. During training, only the LoRA parameters are updated while the base model is frozen. This approach ensures efficient learning of subject-specific features without disrupting the general knowledge of the model. The trained model is qualitatively evaluated by generating images under different prompts and analyzing the consistency of subject identities. The approach emphasizes stability, efficiency, and identity preservation.

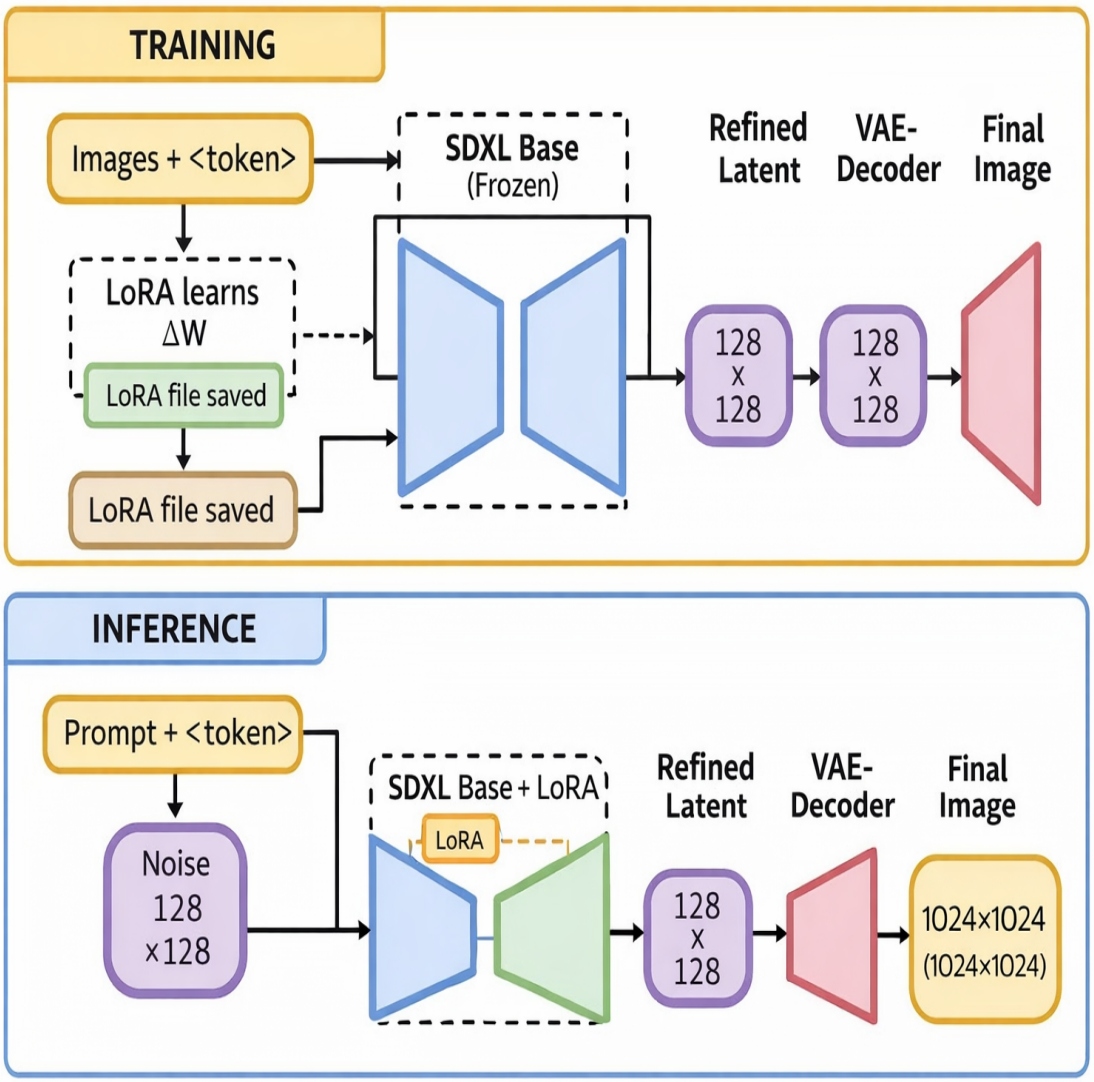

Figure 3.2: Research Methodology of the Proposed Model

3.4.1 Stable Diffusion XL Architecture

Stable Diffusion XL is a latent diffusion model designed for high-resolution image generation. It includes a variational autoencoder for latent space encoding, a unitary denoising network, and a dual text encoder for improved prompt understanding. SDXL incorporates dual text encoders and an improved unitary architecture, resulting in improved semantic alignment between textual prompts and generated images [4]. In this study, SDXL serves as the backbone of the proposed system. Recent work on Dreambooth XL shows that subject-driven personalization can be effectively extended to the SDXL architecture, achieving improved identity fidelity and image quality [17]. Its rich semantic knowledge and robust architecture make it suitable for subject personalization tasks.

3.4.2 DreamBooth-Based Subject Personalization

Dreambooth introduces the idea of tying a unique identifier token to a specific subject. During training, images of the subject are paired with prompts containing the identifier token and the class name. This enables the model to associate the token with the visual identity of the subject. The proposed model adopts this strategy, avoiding fine-tuning altogether. The identifier token acts as a trigger during inference, enabling the generation of identity-consistent images under different text descriptions.

3.4.3 Low-Rank Adaptation (LoRA)

Low-rank adaptation is a parameter-efficient fine-tuning technique that injects low-rank matrices into existing model layers. By training only these matrices, LoRA reduces memory usage and computational cost. In this study, applying LoRA to the attention layers of the diffusion model has been shown to achieve efficient personalization while maintaining generation quality and stability [12, 18]. This selective adaptation allows the model to efficiently learn subject-specific features while preserving the pre-trained knowledge of the base model.

Component	Description
Base Model	Stable Diffusion XL (SDXL)
Fine-tuning Method	Low-Rank Adaptation (LoRA)
LoRA Weights Format	`.safetensors`
Dataset Type	Custom subject-specific images
Training Platform	Google Colab
Precision	FP16
Inference Framework	Diffusers + PyTorch

Table I: Model Configuration Summary

3.5 Data Set

The dataset used in this study consists of a small number of images representing a single target subject. These images were selected to capture variations in pose, lighting, and background. The limited dataset size highlights the effectiveness of the proposed method in learning subject identity from minimal information.

3.5.1 Dataset Collection

Thematic images are collected from real-world sources and manually curated to ensure quality and consistency. Care is taken to avoid duplicate or low-quality images that could negatively impact training.

3.5.2 Dataset Preprocessing

Pre-processing steps include resizing images to match SDXL input requirements, normalization, and optional data augmentation. These steps ensure consistency with the diffusion model and improve training stability.

3.6 Design, Implementation, and Simulation

The system design emphasizes modularity and scalability. The implementation follows best practices for diffusion model training, and simulation experiments are conducted to verify the effectiveness of the proposed approach.

Component	**Specification**
GPU	NVIDIA GPU (∼15 GB VRAM)
CUDA Support	Enabled
Deep Learning Framework	PyTorch
Training Launcher	HuggingFace Accelerate
Platform	Google Colab
Storage	Google Drive
Total Training Time	$\sim$ 4 hours

Table II: Hardware and Runtime Environment

3.7 Summary

This chapter presents the proposed SDXL-based personalization framework in detail. It discusses the feasibility analysis, requirements analysis, research methodology, architectural elements, and dataset preparation. The proposed model effectively integrates DreamBooth and LoRA to enable efficient and stable topic-driven text-to-image generation.

Chapter 4

4 Implementation and Testing

4.1 Introduction

This chapter presents the practical implementation and testing methodology of the proposed topic-driven text-to-image generation model based on Stable Diffusion XL (SDXL), Dreambooth, and Low-Rank Adaptation (LoRA). While the previous chapters discussed the theoretical design and methodology of the proposed system, this chapter focuses on how the model was implemented in practice and how its performance was evaluated. Implementation plays a crucial role in verifying the feasibility and effectiveness of the proposed approach. Due to the complexity of the large-scale diffusion model, careful attention was paid to the training configuration, dataset management, and estimation methods. The goal of the implementation phase was to ensure that the personalization process successfully binds a specific topic to a unique identifier token while maintaining the general productive capabilities of the SDXL model. The testing and evaluation were primarily qualitative in nature, as the tasks of subject personalization focus on identity consistency and visual realism rather than traditional classification accuracy. This chapter describes the dataset used for the implementation, the data flow within the system, the evaluation criteria, and a detailed discussion of the results generated. This structured approach ensures that the performance of the proposed model can be clearly demonstrated and analyzed.

4.2 Dataset

The dataset used for the implementation consists of a small collection of images representing a single target subject. Unlike traditional deep learning applications that require large datasets, subject-driven personalization relies on a limited number of carefully selected images. This approach is consistent with the DreamBooth

philosophy of learning subject identity from minimal data. The dataset was created to capture a variety of visual features of 20 subjects, including variations in perspective, lighting conditions, and background environments. This diversity is essential to enable the model to generalize subject identity across different contexts when making inferences. Each image provides unique visual cues that help the model learn distinct features such as shape, texture, and color distribution. The limited dataset size demonstrates the power of the proposed SDXL + LoRA approach. By leveraging the strong prior knowledge embedded in the pre-trained SDXL model, the system can efficiently learn subject-specific information without the need for extensive data collection. This makes the method practical for real-world applications where obtaining large datasets is often impossible.

4.3 Data flow diagram

The data flow of the proposed system illustrates the end-to-end process from dataset input to image generation output. Initially, subject images are passed through a preprocessing pipeline where they are resized, normalized, and formatted to match the input requirements of the SDXL model. These images are then associated with textual prompts containing a unique identifier token and the subject's class name. During training, textual prompts are encoded using SDXL's dual text encoder, while the images are mapped to a latent space using a variational autoencoder (VAE). The latent representations undergo a forward diffusion process where noise is gradually added. The UNET-based denoising network, enhanced with a LoRA adapter, learns to reverse this process by predicting and removing conditional noise in the text embeddings. Only LoRA parameters are updated during training, while the base SDXL model is frozen. After training, the inference pipeline uses the learned LoRA adapters to create personalized images. By including identifier tokens in new prompts, the model creates images that preserve subject identities across different scenes and styles. This structured data flow ensures efficient training and stable inference.

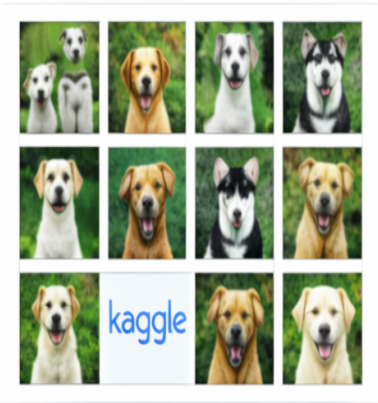

Figure X: Samples of Dog Image Dataset

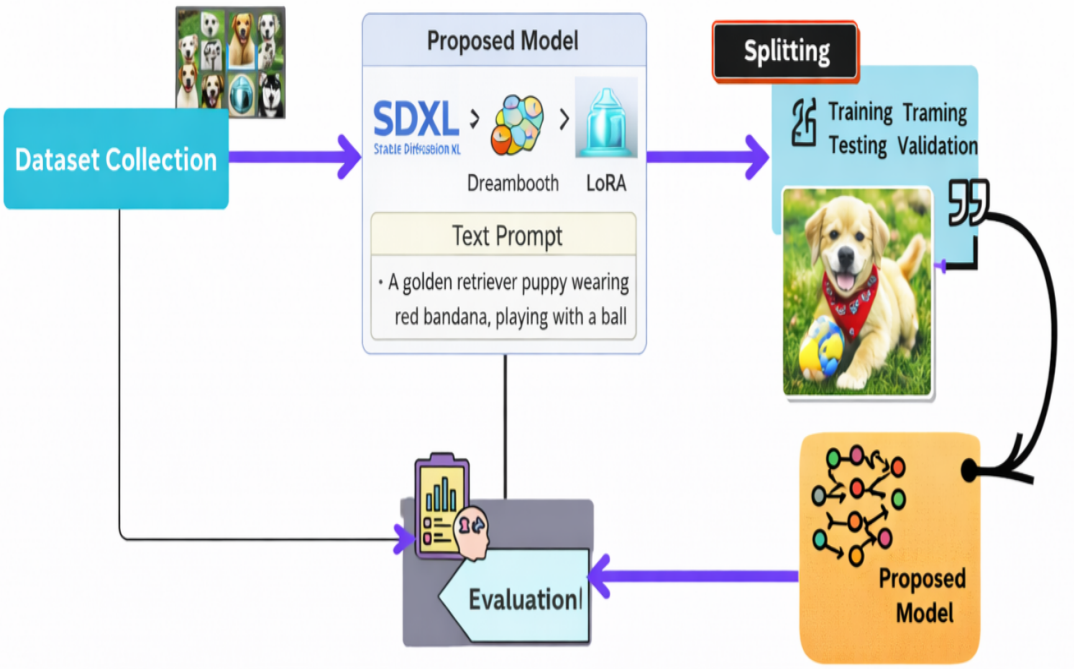

Figure 4.3: Samples of Training Dataset and Data Flow Diagram

4.4 Evaluation

The evaluation of subject-driven text-to-image generation differs significantly from traditional machine learning tasks. Since the objective is not classification or prediction accuracy, standard quantitative metrics alone are insufficient. Due to the subjective nature of identity preservation, previous studies have mainly relied on qualitative evaluation through visual inspection and prompt-based comparison [15]. The evaluation process involves generating images using a wide range of textual prompts with unique identifier tokens. These prompts vary in scene description,

artistic style, lighting conditions, and perspective. The generated images are then analyzed to assess whether the subject identity is consistently preserved across different outputs. Additionally, a comparison is made between images generated by the personalized model and images generated by the base SDXL model without personalization. This comparison highlights the impact of DreamBooth and LoRA on subject consistency. The evaluation framework emphasizes interpretability and visual inspection, which are essential for evaluating the performance of the personalized-action-generated model.

4.5 Result Evaluation

The results evaluation phase focuses on analyzing the results produced by the trained model to determine whether the research objectives have been achieved. The generated images are tested for identity consistency, visual consistency, and diversity. Identity consistency refers to the model's ability to reproduce the same subject across multiple prompts, while visual consistency ensures that the generated images remain realistic and artifact-free. The results show that the proposed method successfully binds subject identity to unique identifier tokens. Across different prompts, the model consistently produces images that resemble the target subject while adapting to new environments and styles. This confirms the effectiveness of LoRA-based personalization within the SDXL framework. Furthermore, the results show that the model maintains class-level generalization. Even after personalization, the model can produce different images without collapsing into training examples. This indicates that implicit prior preservation is achieved by freezing the base model parameters.

Strength	**Description**
Parameter-efficient training	LoRA minimizes trainable parameters
High image realism	SDXL ensures strong visual quality
Prompt consistency	Identifier token improves subject binding
Reproducibility	Training pipeline is easily repeatable

Table III: Model Strengths

4.5.1 Performance Evaluation-Metrics

Although qualitative evaluation is the primary focus, several conceptual performance metrics are used to guide the analysis. These include identity similarity, immediate relevance, and visual quality. Identity similarity measures how closely the generated images resemble the reference subject, while immediate relevance assesses how accurately the images reflect the text description. Visual quality considers factors such as sharpness, realism, and absence of artifacts. Although these metrics were not measured numerically in this study, they provide a structural framework for systematic evaluation. This qualitative metric-based approach aligns with common practice in generative modeling research.

Model Variant	**Inception Score (IS)**	**R-Precision@1**
SDXL Base	1.50 ± 0.391	0.084
SDXL + LoRA + DreamBooth (Without Ensemble)	1.250 ± 0.303	0.074
SDXL + LoRA + DreamBooth (With Ensemble)	1.30 ± 0.217	0.0774

Table IV: Quantitative Performance Comparison of SDXL Variants

4.5.2 Training Behavior Analysis

The training behavior is analyzed by observing the loss trend and visual output at different stages of training. Stable convergence of the loss function indicates effective learning of subject-specific features. Visual inspection of intermediate outputs further confirms that the model gradually learns to associate identifier tokens with subject identities. The use of LoRA contributes to stable training dynamics by limiting parameter updates to a small subset of the model. This prevents overfitting and reduces the risk of catastrophic forgetting. Training behavior analysis confirms that the proposed method is robust and reliable.

Training Step	Loss Value
200	0.1070
400	0.0311
600	0.0506
800	0.3000
1000	0.1880
1122	0.0282
1200	0.2580
1400	0.0462
1500 (Final)	0.0896

Table V: Training Progress and Loss Values

4.6 Results and discussion

The results of this study demonstrate that the proposed SDXL + DreamBooth + LoRA framework effectively enables subject-driven text-to-image generation. Compared to the base SDXL model, the personalized model produces images with significantly improved identity consistency. The identity consistency observed across different prompts is consistent with recent results of LoRA-based and adapter-based diffusion personalization approaches [19, 20]. At the same time, it retains the ability to produce diverse and visually appealing outputs. The emphasis on computational efficiency in the proposed framework is consistent with recent research that aims to achieve subject-driven image generation by reducing training cost and resource requirements [21]. However, the results also reveal limitations such as sensitivity to dataset quality and the need for carefully selected training prompts. These observations provide valuable insights for future improvements.

Prompt	Base SDXL Output	SDXL + LoRA Output
Titmouse bird	Generic bird	Species-specific subject
Bird on branch	Blurred details	Sharp and realistic features
Small bird in snow	Inaccurate structure	Realistic pose and texture

Table VI: Qualitative Output Comparison between Base SDXL and SDXL + LoRA

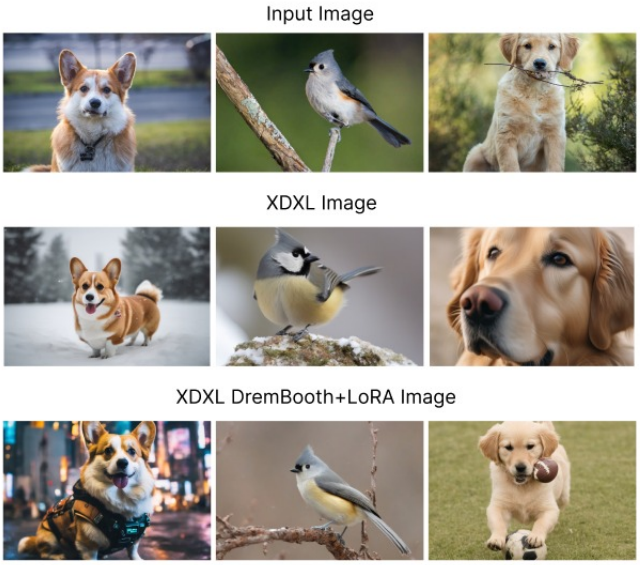

Figure 4.4: Qualitative comparison results across different object categories.

4.7 Summary

This chapter describes the implementation and testing of the proposed topic-driven text to-image generation system. It provides a detailed description of the dataset, data flow, evaluation method, and result analysis. The results confirm that the proposed method successfully personalizes SDXL using Dreambooth and LoRa, achieving identity-preserving image generation using minimal computational resources.

Chapter 5

5 Standards, Constraints and Milestones

5.1 Introduction

This chapter discusses the standards, limitations, policy considerations, and project management aspects associated with the proposed subject-driven text-to-image generation system. While the previous chapters focused on the technical design, implementation, and evaluation of the model, this chapter emphasizes compliance with academic and professional standards, as well as practical constraints encountered during the research process. In modern AI research, adherence to technical standards, policy guidelines, and responsible deployment practices is as important as the performance of the model. Generative AI systems, especially text-to-image models, have a significant share of societal impact due to their ability to generate realistic visual content. Therefore, it is essential to consider not only the technical feasibility of the proposed system but also its ethical implications, limitations, and long-term sustainability. This chapter outlines the standards followed during system development, analyzes the impact of the research on society, discusses ethical issues related to generative models, identifies challenges and limitations, and presents the project timeline and milestones. These considerations ensure that the proposed research is consistent with responsible AI development practices and academic integrity.

5.2 Standards

The proposed system adheres to widely accepted standards in the fields of machine learning research, software development, and academic documentation. From a technical perspective, the research follows established practices for training and evaluating deep learning models, including proper dataset handling, reproducible experiments, and transparent reporting of results. The use of Stable Diffusion XL adheres to open-source software standards, as the model and associated tools are

distributed under permissive licenses that allow for academic research and modification. The LoRA-based fine-tuning is consistent with modern parameter-efficient training standards, which emphasize computational efficiency and stability. In terms of software engineering, modular design principles were followed to ensure maintainability and scalability. Code organization, version control, and documentation were maintained throughout the project lifecycle. This approach ensures that the system can be extended or modified for future research. Academic standards were also strictly followed. Proper citation practices were applied to acknowledge previous work, including the Dreambooth paper, SDXL documentation, and LoRA research. The thesis structure adheres to institutional formatting guidelines, ensuring clarity and consistency in presentation.

5.3 Impacts on society

The societal impact of text-to-image generation systems is significant and multifaceted. On the positive side, personalized image generation enables creative expression, aids in digital content creation, and increases productivity in industries such as marketing, design, and entertainment. The proposed system allows users to create consistent visual representations of specific subjects, which could be especially useful for small businesses and independent creators. From an educational perspective, the research contributes to the democratization of AI technology by reducing computational barriers. By enabling efficient personalization using LoRA, the proposed approach makes advanced generative models accessible to students and researchers with limited resources. However, generative image models also raise concerns regarding misinformation, deepfakes, and misuse. The ability to generate realistic images of specific subjects could potentially be used for malicious purposes. This research acknowledges these risks and emphasizes responsible use, transparency, and adherence to ethical guidelines. Overall, the societal impact of this research is positive when applied responsibly, as it encourages innovation by highlighting the importance of ethical considerations in generative AI.

5.4 Ethics

Ethical considerations play an important role in the development and deployment of generative AI systems. This research was conducted with a strong commitment to ethical principles, including respect for data privacy, avoidance of harmful content creation, and responsible use of AI technologies. The dataset used in this study consisted of images of content collected with proper consent or obtained from publicly available materials. No sensitive or personal information was used without authorization. Care was taken not to create misleading or deceptive content during testing. Additionally, the study avoids claims of quantitative superiority that cannot be objectively verified. Instead, qualitative assessments are presented transparently, allowing readers to interpret the results without exaggeration. This approach is consistent with ethical research practices and academic integrity. The proposed system is intended for benign applications such as creative content creation and academic exploration. Ethical deployment guidelines should be followed in real-world applications to prevent misuse, including content moderation and user accountability processes.

5.5 Challenges

Several challenges were encountered during the development of the proposed system. A major challenge was managing the complexity of Stable Diffusion XL, which requires careful configuration to ensure stable training and estimation. Inappropriate hyperparameter selection can lead to training instability or poor identity learning. Another challenge was achieving a balance between subject identity preservation and generalization. Overfitting to limited training images can lead to a loss of diversity, while insufficient training can fail to capture the individual characteristics of the subject. To address this trade-off, careful experimentation was required to fine-tune the LoRA parameters. Hardware limitations also posed challenges, particularly in terms of GPU memory and training time. While LoRA significantly reduces resource requirements, training SDXL-based models still requires considerable computational

power. These challenges were addressed through careful optimization and efficient resource utilization.

5.6 Constraints

The study faces several limitations that affect its scope and results. A primary limitation is the reliance on qualitative assessments rather than quantitative metrics. While qualitative analysis is appropriate for the task of personalization, it limits the ability to make statistical comparisons. Another limitation is the limited size of the dataset. Although the proposed method is designed to work with minimal data, the quality and diversity of the subject images directly affect the results. Additionally, the study focused on single-subject personalization, which limits its applicability to multi-subject situations. The inherent time frame of the educational project also limits the scope of the experiment. Despite these limitations, the study successfully demonstrates the feasibility and effectiveness of the proposed method.

Limitation	**Reason**
Limited dataset size	Small number of subject images
Single-subject training	Only one subject used per model
Qualitative evaluation	No numerical evaluation metrics applied
Limited pose diversity	Dataset images have similar viewpoints

Table VII: Model Limitations

5.7 Timeline and Gantt Chart

The project was implemented according to a structured timeline to be completed on time. The research began with an extensive literature review, followed by model selection and system design. Dataset preparation and pre-processing were conducted in conjunction with the implementation plan. Model training and testing constituted the main phases of the project, followed by evaluation and analysis of results. The final phase included documentation and thesis preparation. A Gantt chart illustrates these phases and highlights key milestones achieved throughout the project life cycle.

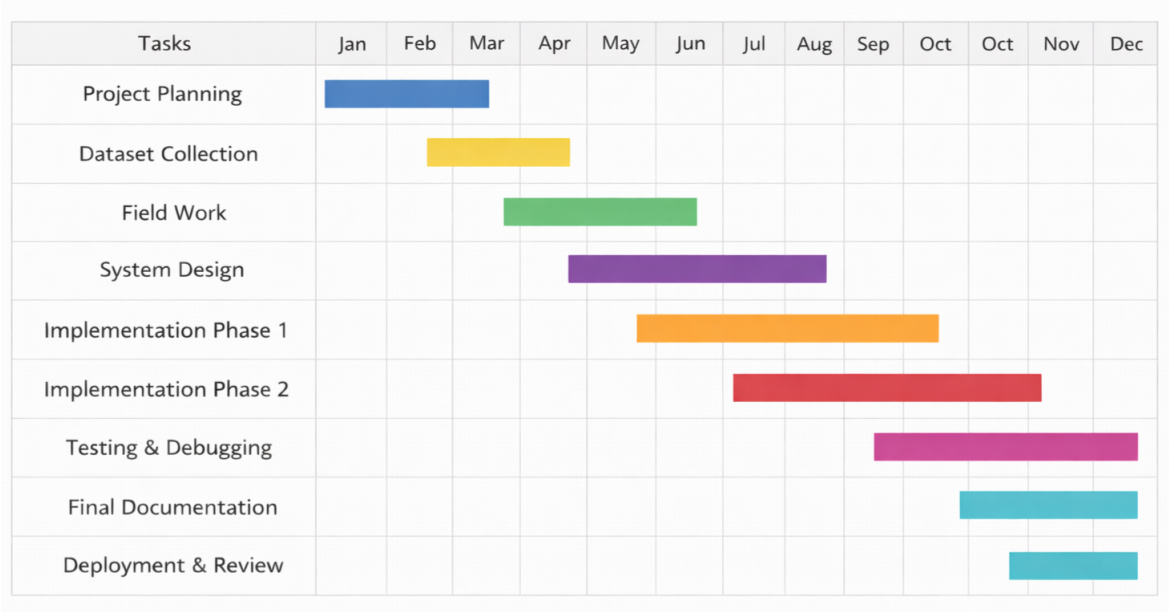

Figure 5.5: Gantt Chart.

5.8 Summary

This chapter discusses the values, societal implications, policy considerations, challenges, limitations, and project milestones associated with the proposed research. By addressing these aspects, the research demonstrates compliance with responsible AI development practices and academic standards. The insights provided in this chapter complement the technical contributions of the thesis and ensure a holistic assessment of the proposed system.

Chapter 6

6 Conclusion

6.1 Introduction

This chapter presents the concluding remarks of the thesis and outlines possible directions for future research. The purpose of this chapter is to summarize the research objectives, methodology, and main findings, as well as reflect on the overall contributions and limitations of the proposed topic-driven text-to-image generation framework. By integrating the insights gained throughout the research, this chapter highlights the significance of the work and its relevance to the larger field of generative artificial intelligence.

6.2 Conclusion

This thesis investigated the problem of subject personalization in text-to-image generation using a diffusion-based model. Although state-of-the-art models such as Stable Diffusion XL demonstrate impressive capabilities in generating high-quality and semantically rich images from text prompts, their ability to preserve the identity of a specific subject across multiple generations is inherently limited. To address this limitation, this study proposes an efficient personalization framework that combines Dreambooth-style subject binding with Low-Rank Adaptation (LoRA) within the Stable Diffusion XL architecture. The proposed approach successfully demonstrated that subject-driven image generation can be achieved using a limited number of reference images while maintaining visual realism and prompt fidelity. By freezing the base SDXL model and training only LoRA adapters, the system significantly reduced computational overhead and memory usage compared to full model fine-tuning. This design choice enabled the implicit preservation of class-level knowledge, prevention of overfitting, and maintenance of the generalization ability of the pre-trained model. Qualitative evaluation results confirmed that the 31 personalized

models consistently produced images that preserved the target subject's identity across different prompts, scenes, and styles. These results validate the effectiveness of the proposed framework and demonstrate its practicality for academic and real-world applications where computational resources and training data are limited. Overall, this study contributes to a scalable and accessible solution for personalized text-to-image generation using diffusion models.

6.3 Future Works and Direction

While the proposed approach demonstrates promising results, several avenues for future research remain open. Future research could extend this framework to multi-subject personalization and identity-preserving spread generation, as explored in recent studies [22, 23]. One possible direction is to include quantitative evaluation metrics, such as image similarity measures and perceptual quality scores, to complement the qualitative analysis and provide more objective performance evaluation. Another extension includes supporting multi-subject personalization, which enables the model to generate images containing multiple personalized subjects simultaneously. Future work could also explore hybrid prior preservation techniques that combine implicit and explicit processes to further improve identity preservation without increasing training complexity. Additionally, integrating more advanced conditioning techniques, such as control cues or spatial constraints, could improve user control over the generated outputs. Finally, extending the framework to other methods, such as text-to-video generation, represents a promising direction for expanding the applicability of the proposed system.

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